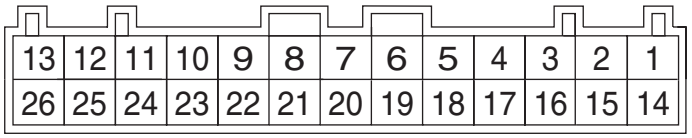
4. Security System

A: WIRING DIAGRAM

<Ref. to WI-171, WIRING DIAGRAM, Security System.>

B: ELECTRICAL SPECIFICATION



SL-00478

Contents	Terminal No.	Measuring condition
Power supply (Backup)	2	Battery voltage is constantly present.
Door switch & rear gate latch switch (wagon model)	6 (INPUT)	Less than 1 V is detected when one of the doors or the rear gate is opened.
Trunk room light switch (sedan model)	7 (INPUT)	Less than 1 V is detected when the trunk lid is opened.
Impact sensor	8	—
Ignition switch (ON)	10 (INPUT)	Battery voltage is present when the ignition switch is turned to ON.
Security horn relay	11 (INPUT)	Battery voltage is detected when the alarm operation is activated.
Interrupt relay	12 (OUTPUT)	Battery voltage is detected when the alarm is activated.
Ground	14	Less than 10 Ω is continually detected.
Impact sensor	15	—
Security indicator light	15 (OUTPUT)	Less than 1 V is detected when the alarm is activated.
Impact sensor	17	—

C: INSPECTION

1. BASIC DIAGNOSTIC PROCEDURE

NOTE:
- Turbo models are equipped with an immobilizer.
- Non-turbo models are not equipped with immobilizers.

	Step	Check	Yes	No
1	**SECURITY SYSTEM SETTINGS.** Turn the security system setting to ON. <Ref. to SL-24, SYMPTOM CHART, INSPECTION, Security System.>	Is the setting process completed correctly?	Go to step **2**.	• Check the ignition switch circuit. <Ref. to SL-28, CHECK IGNITION SWITCH CIRCUIT, INSPECTION, Security System.> • Check the door lock switch circuit. <Ref. to SL-9, CHECK DOOR LOCK SWITCH AND CIRCUIT, INSPECTION, Door Lock Control System.>
2	**CHECK SECURITY SYSTEM SETTING OPERATION.** 1) Before starting this check, open all doors. 2) Pull out the key from the ignition key cylinder, and close all doors and the rear gate. 3) Press the LOCK/ARM button of the transmitter.	Can the security system be set?	Go to step **3**.	Go to symptom 1. <Ref. to SL-24, SYMPTOM CHART, INSPECTION, Security System.>
3	**CHECK SECURITY INDICATOR AND HAZARD LIGHT FLASHING CONDITION.** Check the security indicator and hazard light flashing condition. NOTE: Models with the immobilizer will flash twice, and those without will flash once.	Does the security indicator and hazard light flash?	Go to step **4**.	Go to symptom 2. <Ref. to SL-24, SYMPTOM CHART, INSPECTION, Security System.>
4	**CHECK SECURITY SYSTEM OPERATION.** Press the LOCK button of keyless transmitter, and wait for 30 seconds.	Does the security indicator light flash twice within 0.5 seconds in two second intervals?	Go to step **5**.	Replace the keyless control module.
5	**CHECK SECURITY ALARM OPERATION.** 1) Unlock all doors using the door lock switch on the front door. 2) Open any door or rear gate.	Does the security alarm operate when opening any door or rear gate?	Go to step **6**.	Go to symptom 3. <Ref. to SL-24, SYMPTOM CHART, INSPECTION, Security System.>
6	**CHECK SECURITY ALARM OPERATION.** Check the security alarm operation.	Do all security alarms (horn, hazard light and security indicator light) operate, and the starter motor stop operation?	Go to step **7**.	Go to symptom 4. <Ref. to SL-24, SYMPTOM CHART, INSPECTION, Security System.>
7	**CHECK SECURITY ALARM CANCEL OPERATION.** Press the UNLOCK/DISARM button of the keyless transmitter. NOTE: Models with the immobilizer will flash once, and those without will remain unlit.	Do all security alarms (horn and hazard light) stop operation, and does the starter motor operate?	Go to step **8**.	Go to symptom 5. <Ref. to SL-24, SYMPTOM CHART, INSPECTION, Security System.>

Step		Check	Yes	No
8	**CHECK BATTERY OPEN CIRCUIT PROTECTION FEATURE.** Check that the system functions properly even if the battery is not connected temporally. <Ref. to SL-23, CHECK BATTERY OPEN CIRCUIT PROTECTION FEATURE, INSPECTION, Security System.>	Does the system function properly when the battery is not connected temporarily?	Go to step **9**.	Replace the keyless control module.
9	**CHECK IMPACT SENSOR.** Check the sensitivity of impact sensor. <Ref. to SL-53, CHECK IMPACT SENSOR, ADJUSTMENT, Impact Sensor.>	Is the sensibility set properly?	Press the UNLOCK/DISARM button of the transmitter, and finish the diagnosis.	Adjust the sensibility. <Ref. to SL-53, IMPACT SENSITIVITY ADJUSTMENT, ADJUSTMENT, Impact Sensor.>

2. CHECK BATTERY OPEN CIRCUIT PROTECTION FEATURE

1) Remove the key from ignition switch.
2) Open the front hood.
3) Close all the doors and the rear gate.
4) Press the LOCK/ARM button of the transmitter.
5) Disconnect the ground cable from the battery.
6) Connect the ground cable to the battery.
7) After connecting the battery negative terminal, check whether or not the security indicator light flashes.
If it does not flash, replace the keyless entry control module.

3. SECURITY SYSTEM ON/OFF SETTING

1) Close all doors, trunk lid and rear gate, and sit down on the driver seat. Press the UNLOCK button of the keyless transmitter.
2) Turn the ignition switch to ON.
3) At the same time as pushing the UNLOCK side of the central door lock switch, open the driver's side door, and maintain this condition for 10 seconds.
4) The security system setting is switched (ON ←→ OFF) and the horn sounds.

Setting	Notice
OFF → ON	The horn sounds once.
ON → OFF	The horn sounds twice.

NOTE:
For information on setting ON/OFF of the security system through the Subaru Select Monitor, refer to the following:<Ref. to SL-49, PROCEDURE, Security Control Unit.>

4. SYMPTOM CHART

Symptom			Repair order	Reference
1	The security system cannot be set.		1. Check the transmitter function.	<Ref. to SL-15, CHECK KEYLESS TRANSMITTER BATTERY AND FUNCTION, INSPECTION, Keyless Entry System.>
			2. Check the fuse.	<Ref. to SL-25, CHECK FUSE, INSPECTION, Security System.>
			3. Check the keyless entry control module power supply and ground circuit.	<Ref. to SL-25, CHECK POWER SUPPLY AND GROUND CIRCUIT, INSPECTION, Security System.>
			4. Check the door switch.	<Ref. to SL-26, CHECK DOOR SWITCH, INSPECTION, Security System.>
			5. Replace the keyless entry control module.	<Ref. to SL-49, Security Control Unit.>
2	The security system can be set but the security indicator or hazard light does not flash.	Security indicator light	Check the security indicator light circuit.	<Ref. to SL-27, CHECK SECURITY INDICATOR LIGHT CIRCUIT, INSPECTION, Security System.>
		Hazard light	Check the hazard light operation.	<Ref. to SL-28, CHECK HAZARD LIGHT OPERATION, INSPECTION, Security System.>
3	When one of the doors, trunk lid or rear gate is opened, the security system alarm does not sound.		Check the door switch.	<Ref. to SL-26, CHECK DOOR SWITCH, INSPECTION, Security System.>
4	The security alarm does not operate.	All functions	Check the door switch.	<Ref. to SL-26, CHECK DOOR SWITCH, INSPECTION, Security System.>
		Security indicator light	Check the security indicator light circuit.	<Ref. to SL-27, CHECK SECURITY INDICATOR LIGHT CIRCUIT, INSPECTION, Security System.>
		Security horn	Check the security horn.	<Ref. to SL-27, CHECK SECURITY HORN, INSPECTION, Security System.>
		Hazard light	Check the hazard light operation.	<Ref. to SL-28, CHECK HAZARD LIGHT OPERATION, INSPECTION, Security System.>
		Starter motor does not operate.	Check the interrupt relay circuit.	<Ref. to SL-28, CHECK INTERRUPT RELAY CIRCUIT, INSPECTION, Security System.>
5	The security system cannot be cancelled.	Transmitter	Check the function of the transmitter.	<Ref. to SL-15, CHECK KEYLESS TRANSMITTER BATTERY AND FUNCTION, INSPECTION, Keyless Entry System.>
		Ignition switch	Check the ignition switch circuit.	<Ref. to SL-28, CHECK IGNITION SWITCH CIRCUIT, INSPECTION, Security System.>

5. CHECK FUSE

Step		Check	Yes	No
1	**CHECK FUSE.** Remove and visually check the fuse No. 2 (in the main fuse box).	Is the fuse blown out?	Replace the fuse with a new part.	Check the power supply and ground circuit. <Ref. to SL-25, CHECK POWER SUPPLY AND GROUND CIRCUIT, INSPECTION, Security System.>

6. CHECK POWER SUPPLY AND GROUND CIRCUIT

Step		Check	Yes	No
1	**CHECK POWER SUPPLY.** 1) Disconnect the harness connector of the keyless entry control module. 2) Measure the voltage between harness connector terminal and chassis ground. ***Connector & terminal*** ***(B176) No. 2 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **2**.	Check the harness for open circuits or shorts between the keyless entry control module and fuse.
2	**CHECK GROUND CIRCUIT.** Measure the resistance between harness connector terminal and chassis ground. ***Connector & terminal*** ***(B176) No. 14 — Chassis ground:***	Is the resistance less than 10 Ω?	The power supply and ground circuit are OK.	Repair the harness.

7. CHECK DOOR SWITCH

	Step	Check	Yes	No
1	**CHECK DOOR SWITCH CIRCUIT.** When one of the doors or the rear gate is opened, measure the voltage between the keyless entry control module harness connector terminal and chassis ground. NOTE: If there is electrical noise, it may not be at 0 V. ***Connector & terminal*** ***Front, rear door and rear gate (wagon model):*** ***(B176) No. 6 (+) — Chassis ground (–):*** ***Trunk lid (sedan model):*** ***(B176) No. 7 (+) — Chassis ground (–):***	Is the voltage 0 V?	Go to step **2**.	Go to step **3**.
2	**CHECK DOOR SWITCH CIRCUIT.** When all doors and the rear gate is closed, measure the voltage between the keyless entry control module harness connector terminal and chassis ground. ***Connector & terminal*** ***Front, rear door and rear gate (wagon model):*** ***(B176) No. 6 (+) — Chassis ground (–):*** ***Trunk lid (sedan model):*** ***(B176) No. 7 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	The door switch is OK.	Go to step **3**.
3	**CHECK DOOR SWITCH.** 1) Disconnect the door switch harness connector. 2) Measure the resistance between door switch terminals. ***Terminals*** ***Door switch No. 1 — No. 3:*** ***Rear gate latch switch (wagon model) No. 1 — No. 2:*** ***Trunk room light switch (sedan model) No. 1 — No. 2:***	Is the resistance 1 MΩ or more when door switch is pushed?	Go to step **4**.	Replace the door switch.
4	**CHECK DOOR SWITCH.** Measure the resistance between door switch terminals. ***Terminals*** ***Door switch No. 1 — No. 3:*** ***Rear gate latch switch (wagon model) No. 1 — No. 2:*** ***Trunk room light switch (sedan model) No. 1 — No. 2:***	Is the resistance less than 1 Ω when door switch is released?	Check the harness for open circuits or shorts between the keyless entry control module and the door switch.	Replace the door switch.

8. CHECK SECURITY INDICATOR LIGHT CIRCUIT

	Step	Check	Yes	No
1	**CHECK SECURITY INDICATOR LIGHT.** 1) Disconnect the harness connector of the keyless entry control module. 2) Connect the harness connector terminal to ground using a suitable lead wire. ***Connector & terminal*** ***(B176) No. 15 — Chassis ground:***	Does the security indicator light illuminate?	Replace the keyless entry control module.	Go to step **2**.
2	**CHECK POWER SUPPLY FOR SECURITY INDICATOR LIGHT.** 1) Disconnect the connector from the combination meter. 2) Measure the voltage between the combination meter harness connector terminal and chassis ground. ***Connector & terminal*** ***(i11) No. 10 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **3**.	Check the harness for open or short circuits between combination meter and fuse.
3	**CHECK SECURITY INDICATOR LIGHT CIRCUIT.** Measure the resistance between the combination meter harness connector terminal and the keyless entry control module harness connector terminal. ***Connector & terminal*** ***(i10) No. 6 — (B176) No. 15:***	Is the resistance less than 10 Ω?	Replace the combination meter.	Check the harness for open or short circuits between the combination meter and keyless entry control module.

9. CHECK SECURITY HORN

	Step	Check	Yes	No
1	**CHECK SECURITY HORN RELAY.** Remove and check the security horn relay. <Ref. to SL-52, Security Horn Relay.>	Is the security horn operating normally?	Go to step **2**.	Replace the security horn relay.
2	**CHECK POWER SUPPLY FOR SECURITY HORN RELAY.** Measure the voltage between the security horn relay harness connector terminal and chassis ground. ***Connector & terminal*** ***(B243) No. 1 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **3**.	Check the harness for open circuits or shorts between the security horn relay and the horn relay.
3	**CHECK POWER SUPPLY FOR SECURITY HORN RELAY.** Measure the voltage between the security horn relay harness connector terminal and chassis ground. ***Connector & terminal*** ***(B243) No. 2 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **4**.	Check the harness for open or short circuits between the security horn relay and the fuse.
4	**CHECK HARNESS BETWEEN SECURITY HORN RELAY AND KEYLESS ENTRY CONTROL MODULE.** Measure the resistance between the security horn relay harness connector terminal and the keyless entry control module harness connector terminal. ***Connector & terminal*** ***(B243) No. 4 — (B176) No. 11:***	Is the resistance less than 10 Ω?	Go to step **5**.	Check the harness for open or short circuits between the security horn relay and the keyless entry control module.
5	**CHECK SECURITY HORN.** Remove and check the security horn. <Ref. to SL-51, Security Horn.>	Is the security horn operating normally?	Replace the keyless entry control module.	Replace the security horn.

10.CHECK HAZARD LIGHT OPERATION

Refer to the procedures for the keyless entry system. <Ref. to SL-20, CHECK HAZARD LIGHT OPERATION, INSPECTION, Keyless Entry System.>

11.CHECK INTERRUPT RELAY CIRCUIT

	Step	Check	Yes	No
1	**CHECK INTERRUPT RELAY.** Remove and check the interrupt relay. <Ref. to SL-54, Interrupt Relay.>	Is the interrupt relay operating properly?	Go to step **2**.	Replace the interrupt relay.
2	**CHECK INTERRUPT RELAY POWER SUPPLY.** Measure the voltage between the interrupt relay harness connector terminal and chassis ground. ***Connector & terminal*** ***(B422) No. 5 (+) — Chassis ground (–):***	When the ignition switch is set to START, is the voltage 10 V or more?	Go to step **3**.	Check the harness for open circuits or shorts between the interrupt relay and the ignition switch.
3	**CHECK HARNESS BETWEEN INTERRUPT RELAY AND KEYLESS ENTRY CONTROL MODULE.** 1) Turn the ignition switch to OFF. 2) Disconnect the harness connector of the keyless entry control module. 3) Measure the voltage between the interrupt relay harness connector terminal and the keyless entry control module harness connector. ***Connector & terminal*** ***(B422) No. 1 — (B176) No. 12:***	Is the resistance less than 10 Ω?	Replace the keyless entry control module.	Check the harness for open circuits or shorts between the interrupt relay and the keyless entry control module.

12.CHECK IGNITION SWITCH CIRCUIT

	Step	Check	Yes	No
1	**CHECK IGNITION SWITCH SIGNAL.** 1) Disconnect the harness connector of the keyless entry control module. 2) Turn the ignition switch to ON. 3) Measure the voltage between harness connector terminal and chassis ground. ***Connector & terminal*** ***(B176) No. 10 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	The ignition switch is normal.	Check the harness for open circuits or shorts between the keyless entry control module and the ignition switch.